V V Satyannarayana Swamy Vasireddi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.13
Sri Chaitanya Junior college	Class XII (BIEAP)	2018	97.30%
Abhyudaya High School	Class X (AP SSC)	2016	9.80%

TECHNICAL SKILLS

- **Hydraulic & Flood Modelling:** TUFLOW HPC, HEC-RAS (1D/2D), MIKE HYDRO River
- **Hydrological Modelling:** HEC-HMS
- **GIS & Spatial Analysis:** QGIS, DEM Processing, Terrain Analysis
- **Programming Languages:** Python, R-programming, MATLAB
- **Hydrologic Analysis:** Flood Frequency Analysis, Streamflow Recession Analysis, Statistical Analysis

RESEAERCH EXPERIENCE

M.Tech (Research Assistant), IIT Hyderabad (Jul 2024- Present)

- Hydrodynamic investigations and Flood Inundation modelling for lower Mahanadi region as apart of the Project "HYDROLOGICAL INVESTIGATIONS OF MAHANADI RIVER BASIN".
- Developing 1D and 2D hydraulic models of the Mahanadi River Basin using TUFLOW HPC and HEC-RAS for flood inundation assessment.

Trainee Engineer, RoadWorks Consultancy Services (Mar 2022- May 2022)

- Assisted in engineering analysis, project documentation and technical reporting.

FLOOD AND HYDRAULIC MODELLING PROJECTS IN RESEARCH

Backwater Effect Analysis due to Reservoir Water Level Variations

- Investigated the influence of reservoir level fluctuations on upstream river hydraulics and water-surface profiles.
- Assessed impacts of downstream boundary conditions on flood propagation and inundation characteristics.

2D Flood Inundation Modelling of the Mahanadi River Basin

- Developed two-dimensional hydrodynamic flood models using TUFLOW HPC.
- Generated flood depth, velocity and inundation maps under multiple flood scenarios.

Streamflow Recession Analysis using Power-Law Relationships

- Analysed streamflow recession characteristics using power-law formulations.
- Estimated recession parameters and investigated catchment response behaviour.

Flood Frequency Analysis and Design Flood Estimation

- Performed statistical analysis of historical streamflow records for design flood estimation.
- Developed return-period flood scenarios for hydraulic modelling and flood risk assessment.

CONSULTANCY PROJECTS

HYDROLOGIC AND HYDRODYNAMIC ASSESMENT FOR A PROJECT SITE

- Assessment of river streamflow near the project site using available open-source hydrological and meteorological data.
- Evaluation of **flood impacts** on the project site under different design flood scenarios.
- Assessment of rainfall-induced **waterlogging** at the project site for different rainfall events.

Part of Hydrodynamic assessment of the **Riverfront** for a city to evaluate flow characteristics, water levels, and flood behaviour under relevant flow conditions.

POSITIONS OF RESPONSIBILITY

- Coordinator, HydroMaster Workshop, IIT Hyderabad
- Placement Coordinator, Department of Civil Engineering, IIT Hyderabad.
- Class Representative for three academic years.
- NCC Sergeant and NCC 'C' Certificate Holder.
- Organized and coordinated departmental technical and student activities.

RELEVENT COURSES

Engineering Hydrology and Hydrologic Systems, Open Channel Hydraulics and Sediment Transport

Remote Sensing and GIS, Advanced Statistical Methods in Hydrology

Advanced computation lab(MATLAB), Computational Fluid Dynamics

Irrigation Engineering, R Programming

Design of Sewage Systems, Groundwater Modelling

EXTRACURRICULAR ACTIVITIES

- Fitness and strength training enthusiast with an interest in running and outdoor activities.
- Passionate about trekking, cycling, travel, and exploring new destinations.
- Enjoy cooking and exploring diverse cuisines.
- Actively participated in social, cultural, and student-led activities.